FAPS: Lightweight Supplementary Results

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Abstract

Failure-Aware Policy Shielding (FAPS) detects impending manipulation failures from SmolVLA activations, gates a residual correction with a calibrated hysteresis trigger, and leaves the base VLA weights unchanged. This handout is a lightweight companion to the poster. It summarizes the run-log evidence we would show publicly before releasing a full paper repository: risk-head ablations, calibration and threshold behavior, layer/PCA diagnostics, residual-flow training evidence, and the current pre-retrain online shield diagnostic.

1 System Overview

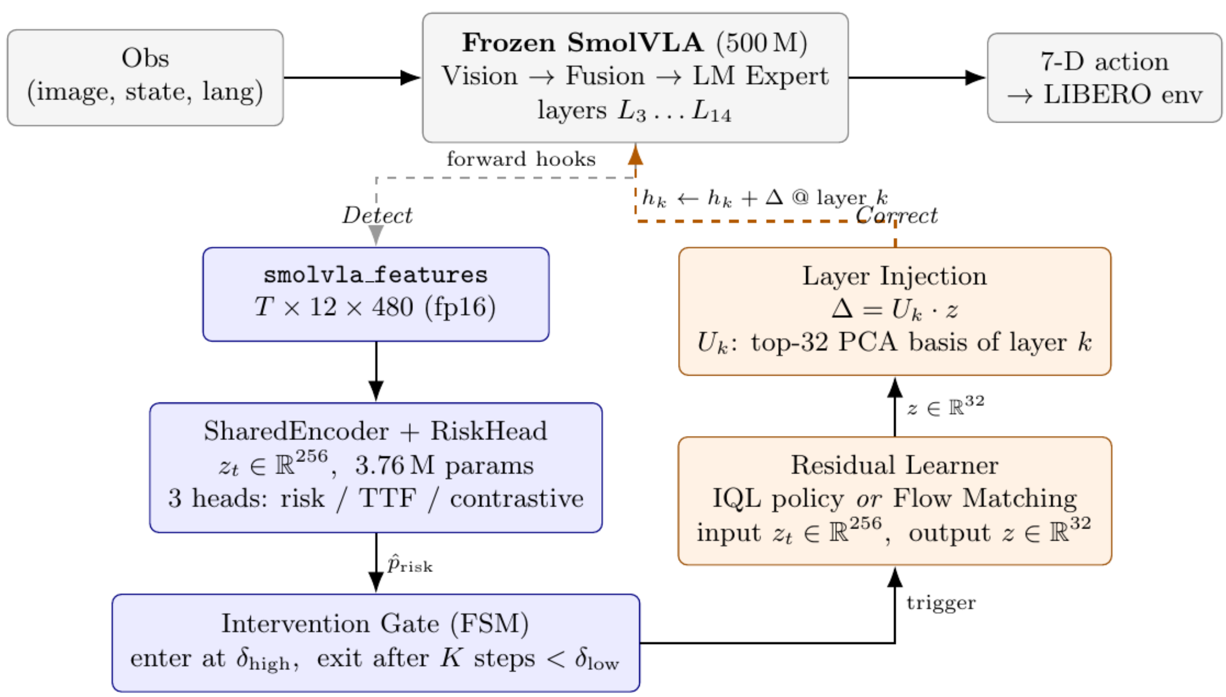

Figure 1: FAPS at inference time. SmolVLA activations feed a risk head; a hysteresis FSM decides when to inject; a flow-matching model proposes a low-dimensional residual that is projected back into the selected layer.

2 Dataset and Labeling

The current labeled corpus contains 5,117 SmolVLA LIBERO rollouts across nominal and perturbation settings. The risk label is `fail_in_H` with horizon $H = 30$, and each failure can also carry a coarse failure type. The planned trigger retrain uses only recoverable/windowed perturbations plus nominal successes, excluding nominal timeout artifacts.

Table 1: Data split used for the immediate retrain plan.

Subset	Count	Intended role
Windowed perturbations	999	Positive and hard local-risk examples
Nominal successes	250	Clean negative examples for trigger precision
Nominal timeout failures	125	Excluded as likely timeout artifacts
Continuous perturbations	4,118	Held out from retrain; later eval/stress set

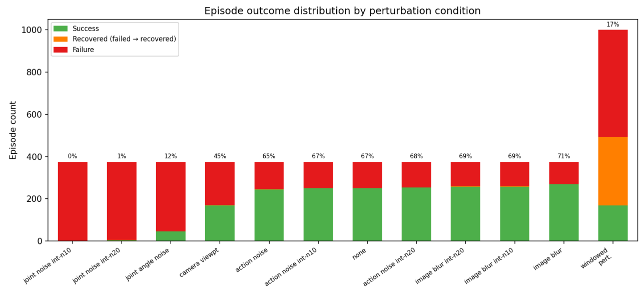

Figure 2: Episode outcome distribution.

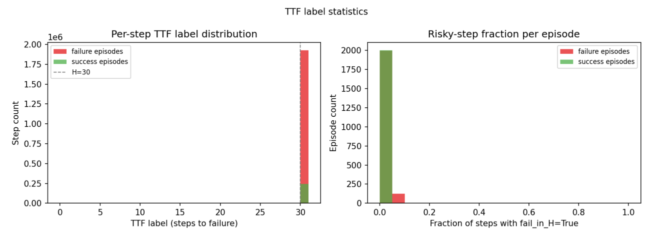

Figure 3: Time-to-failure label distribution.

3 Risk-Head Ablation

The most useful poster table is the A/B/C/D ablation. It shows that BCE is needed for detection, TTF improves temporal structure modestly, and NT-Xent adds useful latent geometry. Run C is selected because it gives the best representation/calibration tradeoff for downstream flow conditioning.

Table 2: Risk-head ablation from completed full/windowed runs.

Run	Losses	wAUC	AP	TTF r	Silh.	ECE	BSkill
A	BCE only	0.928	0.448	0.380	-0.046	0.088	-0.067
B	BCE + TTF	0.927	0.444	0.368	-0.040	0.093	-0.121
C	BCE + TTF + NT-Xent	0.917	0.461	0.413	+0.168	0.077	+0.018
D	NT-Xent only	0.440	0.086	0.075	+0.125	0.319	-1.264

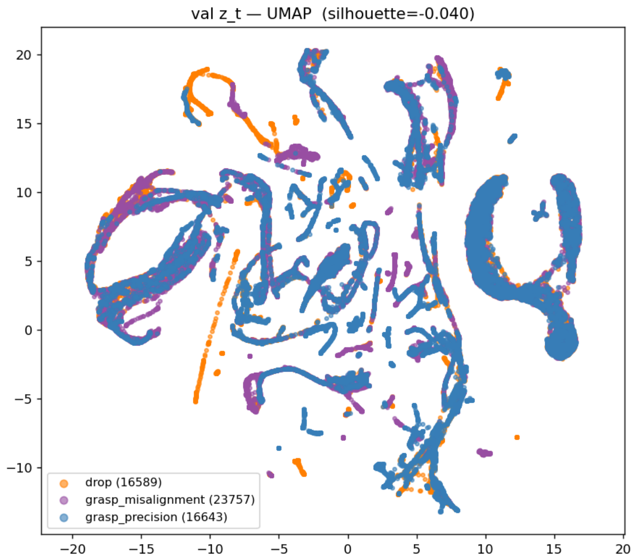

Figure 4: Run B latent UMAP. Detection is strong, geometry is weak.

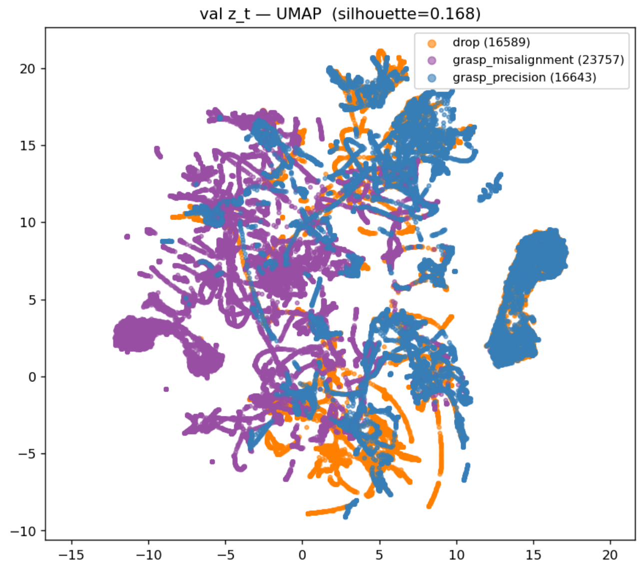

Figure 5: Run C latent UMAP. NT-Xent creates failure-type structure.

4 Calibration and Trigger Sweep

Calibration improves both Run B and Run C. Run C is more attractive for the shield because it uses a stricter high threshold, has a lower intervention rate, and lowers false alarms in the threshold sweep.

Table 3: Temperature calibration and threshold-sweep summary.

Run	Temp.	ECE pre	ECE post	Brier post	δ_{hi}	Interv.	False alarm
B	1.780	0.060	0.052	0.067	0.6	0.114	0.967
C	1.790	0.061	0.053	0.068	0.8	0.065	0.800

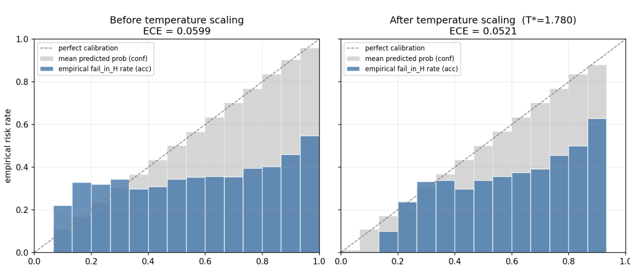

Figure 6: Run B reliability diagram.

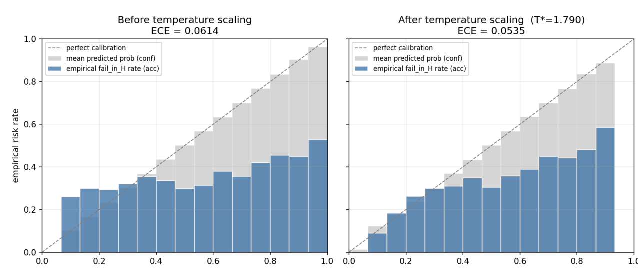

Figure 7: Run C reliability diagram.

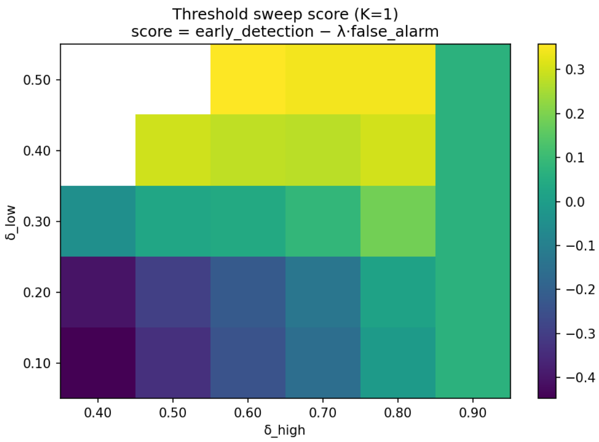

Figure 8: Run B threshold sweep.

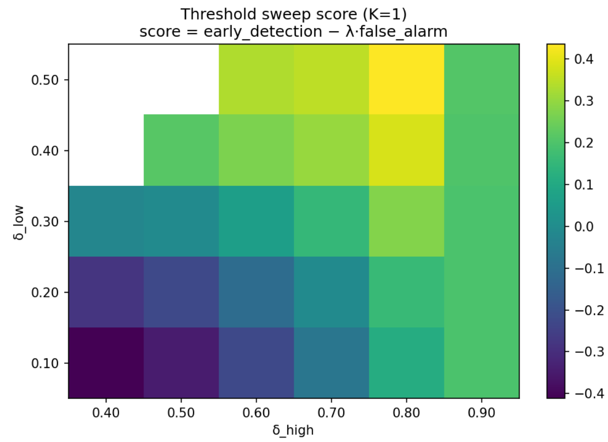

Figure 9: Run C threshold sweep.

5 Layer and PCA Diagnostics

Failure is linearly separable across action-expert layers, so layer choice is a closed-loop control question rather than a pure classification question. PCA diagnostics show that a low-dimensional injection basis is plausible.

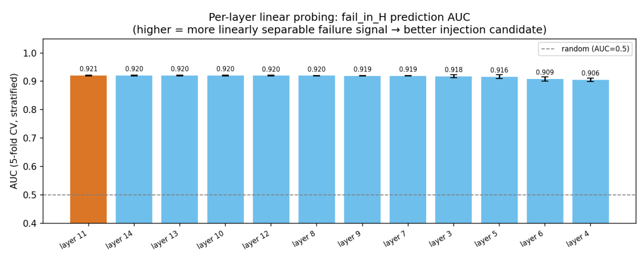

Figure 10: Layer probe AUC. Risk signal is broadly accessible.

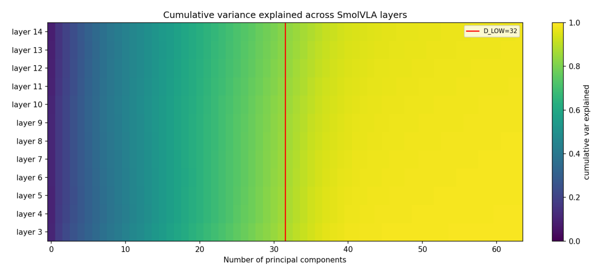

Figure 11: Explained variance heatmap across layers.

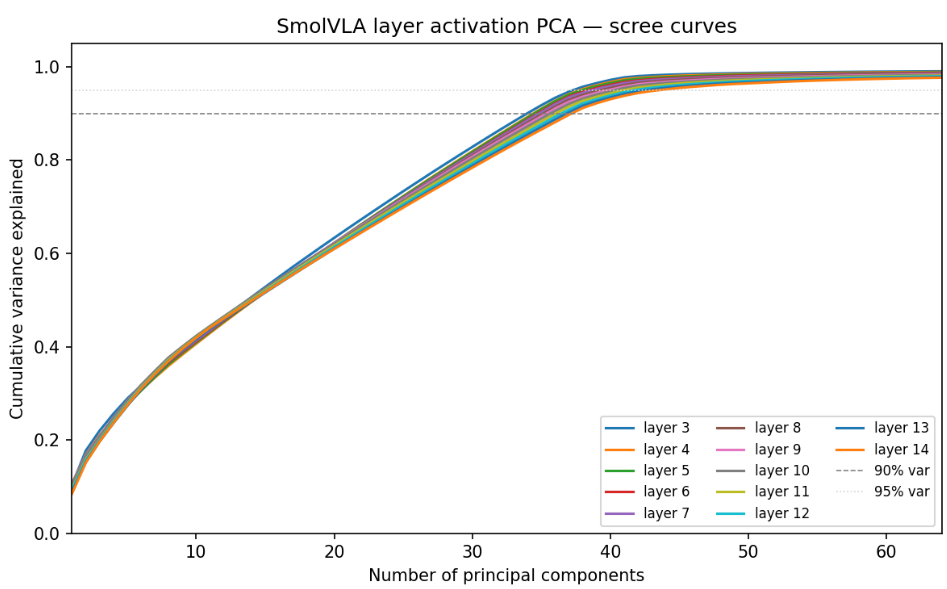

Figure 12: PCA scree curves. The online injector uses a compact top-48 basis for the selected layer.

6 Residual Flow Evidence

Phase-2 flow runs indicate that C12/C13 conceptor+failure conditioning produces substantial residuals, while control runs at layer 11 produce much smaller residuals. This motivates finishing the C12/C13 closed-loop injection studies after the trigger retrain.

Table 4: Flow-matching residual runs.

Run	Layer	Loss	$\|\Delta\|$	Nonzero frac.	Role
flow_c4_l12	12	0.300	1.520	0.930	C12 candidate
flow_c5_l13	13	0.303	1.553	0.930	C13 candidate
flow_b5_failure_conceptor_60k	11	0.312	2.000	0.961	strong residual diagnostic
flow_d_gaussian_l11	11	0.051	0.024	0.758	Gaussian control
flow_z_ownenc_l11	11	0.035	0.082	0.820	own-encoder control

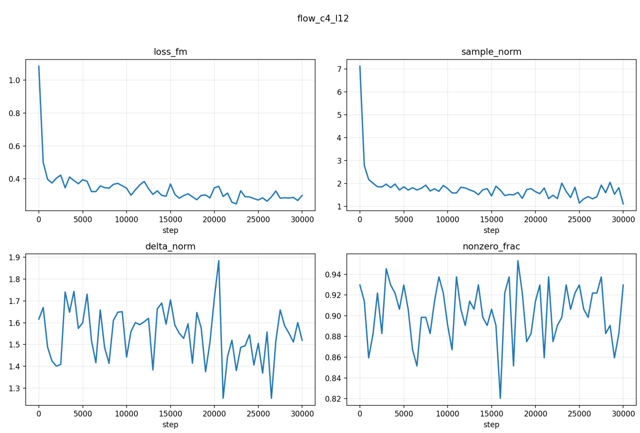

Figure 13: C12 flow training metrics.

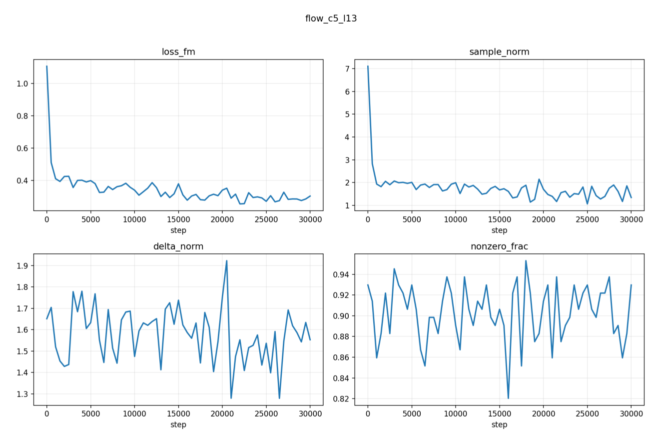

Figure 14: C13 flow training metrics.

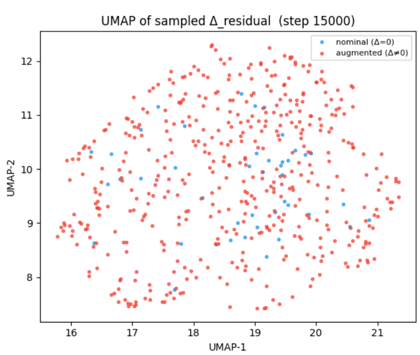

Figure 15: C12 residual UMAP.

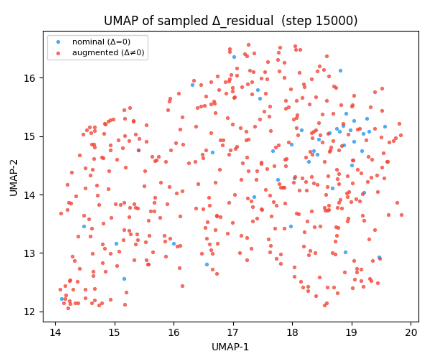

Figure 16: C13 residual UMAP.

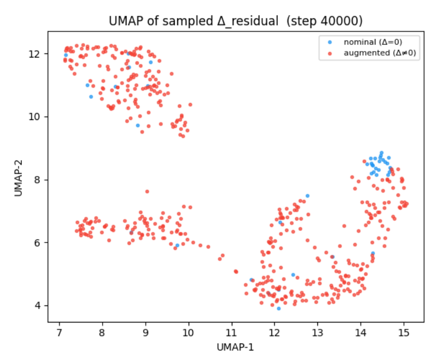

Figure 17: B5 residual UMAP.

7 Pre-Retrain Online Shield Diagnostic

The completed pre-retrain online shield run should be framed as a diagnostic, not the final phase-3 result. It shows that the trigger is currently the weak link: when the trigger avoids false positives, shielded perturbed episodes can succeed; false-positive injection hurts clean or recoverable states.

Table 5: Pre-retrain shield eval: 80 closed-loop LIBERO episodes.

Perturbation	Unshielded SR	FAPS SR	ΔSR	Precision	FPR
none	0.60	0.60	0.00	–	–
action noise	0.70	0.80	+0.10	0.183	0.896
image blur	0.80	0.80	0.00	0.165	0.936
joint noise	0.80	0.70	-0.10	0.104	0.782

Key diagnostic: perturbed shielded episodes with zero false-positive injections succeeded in 17/17 cases; episodes with at least one false-positive injection succeeded only 46%. The immediate fix is to retrain the risk head with nominal successes as clean negative examples and then rerun C12/C13.

8 Next Experimental Step

The next risk-head command should train on windowed perturbations plus nominal successes:

```
modal run --detach modal_train_full.py::train_full \
  --windowed-only --include-nominal --exclude-nominal-failures \
  --epochs 30 --batch-size 64 \
  --w-risk 0.4 --w-ttf 0.2 --w-con 0.4 \
  --contrastive-labels compound --recovery-soft-label 0.3 \
  --windowed-weight 1.0 \
  --wandb-entity cs224r-faps --wandb-project retrain \
  --wandb-group risk_head_retrain --seed 0
```

After retrain, the poster should replace the pre-retrain shield table with closed-loop C12/C13 results and keep the old table as a limitation/diagnostic if space allows.

9 References

This lightweight package is meant to accompany the course poster rather than serve as a full paper. Related areas include activation steering, conceptor steering, VLA robustness under perceptual/action perturbations, and flow-matching residual control.